

The Top-Degree Mass of Unit-Distance Graphs and the Price of Node Privacy

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Abstract

Erdős asked how many pairs among n points in the plane can be at distance exactly 1. We study an unbalanced version of this question and show that it determines the cost of a basic task in data privacy. Let $\Lambda_k(n)$ be the largest possible sum of the k largest degrees in the unit-distance graph of at most n points. We prove that the smallest expected error with which the number of unit-distance pairs can be released under ε -node differential privacy is $\Theta(\Lambda_{\lceil 1/\varepsilon \rceil}(n))$, with absolute constants, for every $\varepsilon \in (0, 1]$. We also show that $\Lambda_k(n)$ equals, up to a factor of 2, the maximum number of incidences between n points and k unit circles. We then compare norms on the plane. $\Lambda_k(n) = \Theta(kn)$ if the unit circle contains a segment; $\Lambda_k(n) = O(n^{2/3}k^{2/3} + n)$ if the norm is strictly convex, and a norm found by Valtr attains this bound for every k ; for almost all norms, $\Lambda_k(n) = \Theta(n)$ up to a logarithmic factor. For the Euclidean norm the optimal error is $\Theta(n)$ when $\varepsilon \geq n^{-1/2}$, a factor $1/\varepsilon$ below the standard Laplace mechanism. It is polynomially larger than for almost all norms if and only if Erdős's unit-distance conjecture fails, which was shown in 2026. Between these ranges the Euclidean question is an open problem on unbalanced unit-circle incidences.

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1 Introduction

Let P be a finite set of points in the plane and let $G(P)$ be its *unit-distance graph*: two points are adjacent if their distance is exactly 1. Erdős [13] asked for the maximum number $u(n)$ of edges of $G(P)$ over n -point sets. The upper bound $u(n) = O(n^{4/3})$ of Spencer, Szemerédi and Trotter [26] has not been improved. Erdős conjectured $u(n) = n^{1+o(1)}$; in 2026 this conjecture was disproved [20, 1], and Sawin [24] gave sets with more than $n^{1.014}$ unit distances for arbitrarily large n .

This paper studies a refinement of $u(n)$ that measures how unevenly unit distances can be distributed among the points. For a finite set P and an integer $k \geq 1$, let $S_k(P)$ be the sum of the k largest degrees of $G(P)$ (padded with zeros if $k > |P|$), and define the *top-degree mass*

$$\Lambda_k(n) = \max\{S_k(P) : |P| \leq n\}.$$

Thus $\Lambda_1(n) = n - 1$ (a point together with $n - 1$ points on its unit circle), and $\Lambda_n(n) = 2u(n)$. In between, $\Lambda_k(n)$ asks how many unit distances k points can see.

The privacy problem. Our motivation comes from differential privacy. Suppose each point of P is the location of one person, and we wish to publish the number $U(P)$ of unit-distance pairs. *Node differential privacy* requires that adding or removing one person, together with all pairs that contain this person, changes the distribution of the published value by at most a factor e^ε (see Section 3 for a self-contained introduction). One point can form a unit-distance pair with all others, so the standard Laplace mechanism must add noise of order n/ε . We



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ask for the *minimax* error

$$\text{err}^*(\varepsilon, n) = \inf_M \sup_{|P| \leq n} \mathbb{E} |M(P) - U(P)|,$$

where M ranges over ε -node-private algorithms and n is a public bound on the number of points. Our first result says that this privacy question is *exactly* the geometric question above.

► **Theorem 1** (Reduction). *For every norm on $\mathbb{R}^2$, every $n \geq 1$ and every $\varepsilon \in (0, 1]$, with $k = \lceil 1/\varepsilon \rceil$,*

$$\frac{1}{8(1+e)} \Lambda_k(n) \leq \text{err}^*(\varepsilon, n) \leq 2 \Lambda_k(n).$$

The upper bound is attained by an algorithm that is ε -node-private on all finite point sets. The same holds, with Λ_k defined accordingly, for every family of point sets closed under deleting points and every graph that is determined by pairs of points (for instance, any distance-bin graph).

The ingredients are known in the privacy literature: a bounded-degree surrogate for the edge count due to Kasiviswanathan, Nissim, Raskhodnikova and Smith [19], and a lower bound by deleting and re-inserting high-degree vertices, which is closely related to the down-sensitivity analysis of Dong, Fang, Yi, Tao and Machanavajjhala [9]. We do not claim these tools as new. What Theorem 1 adds is that, over a family closed under deletion, the worst-case cost is given by a single extremal function. This turns the privacy question into a question about point sets, which is the subject of the rest of the paper.

Geometry of the top-degree mass. The k largest degrees of $G(P)$ count incidences between P and the unit circles centred at k points of P . Let $I(n, k)$ be the maximum number of incidences between at most n points and k distinct unit circles (with arbitrary centres). Equivalently, $I(n, k)$ is the maximum number of unit distances between a set of n points and a set of k points.

► **Theorem 2** (Top-degree mass and incidences). *For every norm on $\mathbb{R}^2$ and all integers $n \geq 2$ and $1 \leq k \leq n$,*

$$\max \left\{ n - 1, \frac{2k u(n)}{n} \right\} \leq \Lambda_k(n) \leq I(n, k),$$

where $u(n)$ is the maximum number of unit distances among n points in this norm. If $k \leq n/2$, then also $I(n, k) \leq 2\Lambda_k(n)$.

Which values can Λ_k take? We answer this across all norms on $\mathbb{R}^2$. A norm is *strictly convex* if its unit circle contains no segment. Valtr [30] found a strictly convex norm in which n points can span $\Omega(n^{4/3})$ unit distances; we show that his norm is extremal for every k , not only for $k = n$. Alon, Bucić and Sauermaann [2] proved that for almost all norms (in the sense of Baire category) every n points span at most $n \log_2 n$ unit distances.

► **Theorem 3** (Top-degree mass across norms). *There are absolute constants $C, c > 0$ such that for all $n \geq 8$ and $1 \leq k \leq n$:*

1. *If the unit circle contains a segment, then $\lfloor n/2 \rfloor \min\{k, n-1\} \leq \Lambda_k(n) \leq (n-1)k$.*
2. *If the norm is strictly convex, then $\Lambda_k(n) \leq C(n^{2/3}k^{2/3} + n)$.*
3. *For Valtr's norm $\|(x, y)\| = |y| + \sqrt{x^2 + y^2}$, we have $\Lambda_k(n) \geq c(n^{2/3}k^{2/3} + n)$.*

Result	Statement	Status
Theorem 1	$\text{err}^*(\varepsilon, n) = \Theta(\Lambda_{\lceil 1/\varepsilon \rceil}(n))$	known tools; new form
Theorem 2	$\Lambda_k(n)$ and $I(n, k)$ agree up to a factor 2	new (short)
Theorem 3(1),(4)	segment norms; almost all norms	new (short); uses [2]
Theorem 3(2)	strictly convex norms	standard incidence bound
Theorem 3(3)	Valtr's norm is extremal for every k	new; extends [30]
Theorem 5	Euclidean window; separation iff $u(n) \geq n^{1+\Omega(1)}$	new connection
Theorem 12	algorithm without knowledge of Λ_k	known tools [19, 23]

■ **Table 1** Results and their provenance.

82 **4.** For almost all norms, $n - 1 \leq \Lambda_k(n) \leq 2n \log_2 n$.

83 Since every norm is either strictly convex or has a segment on its unit circle, Theorems 1
84 and 3 describe the worst possible privacy cost over all norms.

85 ► **Corollary 4** (Privacy cost across norms). For $n \geq 8$ and $1/(n - 1) \leq \varepsilon \leq 1$, the minimax
86 error $\text{err}^*(\varepsilon, n)$ is $\Theta(n/\varepsilon)$ if the unit circle contains a segment, so the Laplace mechanism is
87 optimal; it is $O(n + n^{2/3}\varepsilon^{-2/3})$ for every strictly convex norm, with equality up to constants
88 for Valtr's norm; and it lies between $\Omega(n)$ and $O(n \log n)$ for almost all norms.

89 **The Euclidean plane.** The Euclidean norm is strictly convex, so its privacy cost lies below
90 Valtr's curve. The averaging bound in Theorem 2 brings in $u(n)$.

91 ► **Theorem 5** (Euclidean plane). There are absolute constants $C, c > 0$ such that for the
92 Euclidean norm, all $n \geq 8$ and all $\varepsilon \in [1/n, 1]$,

$$93 \quad c \left(n + \frac{u(n)}{\varepsilon n} \right) \leq \text{err}^*(\varepsilon, n) \leq C \left(n + n^{2/3} \varepsilon^{-2/3} \right).$$

94 In particular $\text{err}^*(\varepsilon, n) = \Theta(n)$ for $\varepsilon \geq n^{-1/2}$. Moreover, the following are equivalent:

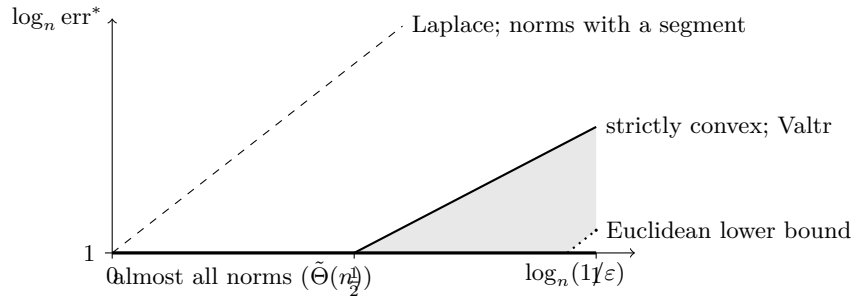
- 95 1. $u(n) \geq n^{1+\delta}$ for some $\delta > 0$ and infinitely many n ;
- 96 2. there are $\delta' > 0$ and privacy levels $\varepsilon_n \in (0, 1]$ such that $\text{err}^*(\varepsilon_n, n) \geq n^{1+\delta'}$ for infinitely
97 many n .

98 By the 2026 results, both hold: for every $\eta \in [0, 0.014)$ there are infinitely many n with
99 $\text{err}^*(n^{-1+\eta}, n) \geq c n^{1.014-\eta}$, while for almost all norms the error is $O(n \log n)$ at every ε .

100 Figure 1 shows the resulting picture. For the Euclidean norm, the region between the two
101 bounds of Theorem 5 is open. By Theorem 2, closing it is the same as determining $I(n, k)$
102 for unit circles when $\sqrt{n} \leq k \leq n$, an unbalanced version of the unit-distance problem. At
103 $k = n$ this is the unit-distance problem itself; Theorem 3(3) shows that any improvement
104 over the incidence bound must use properties of the Euclidean norm that Valtr's norm does
105 not have. We state the question in Section 8.

106 **Summary and provenance.** Table 1 lists the results and says which parts are new. We
107 regard the geometric results (Theorems 2, 3, and 5) as the main contribution. Theorem 1 is
108 the bridge; Section 7 adds an algorithm that does not need to know Λ_k , at a logarithmic cost.

109 **Related work.** *Unit distances and incidences.* Besides the bounds above, the unit-distance
110 problem is surveyed by Brass, Moser and Pach [5]. The $O(n^{4/3})$ bound follows from incidence
111 bounds for points and curves [29, 7, 21], and Székely's crossing-number argument [28] gives it



■ **Figure 1** Exponent of the optimal error against $\log_n(1/\varepsilon)$. Shaded: the range allowed for the Euclidean norm by Theorem 5; its lower edge is drawn with $\delta = 0.06$, while the best known value is $\delta \geq 0.014$.

for any family of curves in which two curves meet in at most two points and two points lie on at most two curves. For normed planes, Swanepoel’s survey [27] describes the known results, including Valtr’s construction. After the disproof of Erdős’s conjecture [20, 1], Sawin [24] gave the exponent 1.014 and Emmerich [12] optimised explicit finite certificates. We use these results only through Theorem 2; we do not use their structure.

Node privacy. Node differential privacy for graphs was studied by Blocki, Blum, Datta and Sheffet [4] and by Kasiviswanathan, Nissim, Raskhodnikova and Smith [19], who introduced the bounded-degree flow statistic we use. For a family closed under deletion, the restricted sensitivity of Blocki, Blum, Datta and Sheffet is $\Lambda_1(n)$ in our notation, which leads to error

about $\Lambda_1(n)/\varepsilon$. Theorem 1 replaces this by $\Lambda_{\lceil 1/\varepsilon \rceil}(n)$, which is never larger; for strictly convex norms it is smaller by a factor of order at least $\varepsilon^{-2/3}$. Raskhodnikova and Smith [23] developed Lipschitz extensions and the generalized exponential mechanism used in Section 7. Instance-dependent guarantees in terms of down sensitivity appear in work of Asi and Duchi [3], Fang, Dong and Yi [14], and Dong, Fang, Yi, Tao and Machanavajjhala [9]; see also Cummings and Durfee [8]. Recent node-private results include counting connected components [18], continual observation [17] and graphon estimation [6]. Raskhodnikova, Smith, Wagaman and Zavyalov [22] study node privacy in the *local* model, where each person randomizes their own edge list; we work in the central model, where a trusted curator holds the data. We are not aware of earlier work that links node privacy to incidence geometry.

2 Overview of the proofs

Why the privacy cost is $\Lambda_{\lceil 1/\varepsilon \rceil}$. The upper bound uses one public threshold. The surrogate g_D of [19] changes by at most D when a point is added or removed, and it undercounts $U(P)$ by at most the degree tail $\text{tail}_D(P) = \sum_p (\deg_P(p) - D)_+$. Set $k = \lceil 1/\varepsilon \rceil$ and $D = \Lambda_k(n)/k$. The vertices of degree above D form a set S , and $\text{tail}_D(P) \leq S_{|S|}(P) - |S|D \leq \Lambda_{|S|}(n) - |S|D$. The average $\Lambda_j(n)/j$ cannot increase with j , so this is at most $\Lambda_k(n)$ for every value of $|S|$. Laplace noise of scale $D/\varepsilon \leq \Lambda_k(n)$ completes the bound.

For the lower bound, take a set P^* with $S_k(P^*) = \Lambda_k(n)$, where now $k \approx 1/\varepsilon$, and remove its k points of largest degree. Re-inserting them one at a time is a path of $k \leq 1/\varepsilon$ neighbouring inputs along which the count grows by at least $\Lambda_k(n)/2$. A private algorithm cannot distinguish the two ends of such a path well, so it errs by $\Omega(\Lambda_k(n))$ on one of them.

Why Λ_k is an incidence count. If C is the set of k points of largest degree in P , then $S_k(P)$ is the number of incidences between P and the unit circles centred at points of C . Conversely, given n points and k unit circles, adding the k centres to the point set turns incidences into degrees. Splitting the points into two halves keeps the total size below n at a loss of a factor 2.

Why the norms differ. If the unit circle contains a segment, two parallel rows of points give a complete bipartite unit-distance graph, so $\Lambda_k(n) = \Theta(kn)$. If the norm is strictly convex, two distinct unit circles meet in at most two points and two points lie on at most two unit circles; the crossing-number argument then gives $I(n, k) = O(n^{2/3}k^{2/3} + n + k)$. For Valtr's norm, the upper half of each unit circle is a parabolic arc, and the map $(x, y) \mapsto (x, y + x^2/2)$ sends all these arcs to segments of lines. The standard grid construction for point-line incidences, at the right aspect ratio, then gives $I(n, k) = \Omega(n^{2/3}k^{2/3})$ for every $\sqrt{n} \leq k \leq n$. For almost all norms the bound of [2] on $u(n)$ caps $\Lambda_k(n) \leq 2u(n)$.

Why the 2026 result enters. The averaging bound $\Lambda_k(n) \geq 2k u(n)/n$ holds because the k largest of n degrees have at least the average degree. With $k = \lceil 1/\varepsilon \rceil$ close to n , Theorem 1 turns a superlinear $u(n)$ into a superlinear privacy cost. Conversely $\Lambda_k(n) \leq \Lambda_n(n) = 2u(n)$, so no other source of superlinear cost exists.

3 Preliminaries

Geometry. Fix a norm $\|\cdot\|$ on $\mathbb{R}^2$ with unit circle $\mathbb{S} = \{x : \|x\| = 1\}$; a *unit circle* is a translate $c + \mathbb{S}$. For a finite set P , $\deg_P(p)$ is the number of $q \in P$ with $\|p - q\| = 1$, and

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$U(P)$ is the number of edges of $G(P)$. For a finite family Γ of curves, $I(P, \Gamma)$ is the number of pairs $(p, \gamma) \in P \times \Gamma$ with $p \in \gamma$. Since $\mathbb{S} = -\mathbb{S}$, a point p lies on $c + \mathbb{S}$ exactly when $\|p - c\| = 1$. We write $\mathcal{P}_n$ for the family of point sets with at most n points. The functions $\Lambda_k(n)$ and $I(n, k)$ are non-decreasing in both arguments, and $\Lambda_k(n) = \Lambda_n(n) = 2u(n)$ for $k \geq n$.

Node differential privacy, for geometers. An algorithm M maps each finite point set P to a random real number $M(P)$. Two point sets are *node neighbours* if one is obtained from the other by adding one point. The word “node” refers to the graph $G(P)$: adding a point adds a vertex together with all its edges.

► **Definition 6.** Let $\varepsilon > 0$ and let $\mathcal{F}$ be a family of finite point sets. An algorithm M is ε -node-private on $\mathcal{F}$ if for all node neighbours $P, Q \in \mathcal{F}$ and every measurable $T \subseteq \mathbb{R}$, $\Pr[M(P) \in T] \leq e^\varepsilon \Pr[M(Q) \in T]$.

The definition says that the output reveals little about whether any single person is present. Smaller ε means stronger privacy; the range $\varepsilon \leq 1$ is the usual one. We use three standard facts [10, 11]. (i) *Laplace mechanism*: if $|f(P) - f(Q)| \leq \Delta$ for all node neighbours, then $f(P) + Z$ with Z drawn from the Laplace density $\frac{\varepsilon}{2\Delta} e^{-\varepsilon|z|/\Delta}$ is ε -node-private, and $\mathbb{E}|Z| = \Delta/\varepsilon$. For $f = U$ one has $\Delta = n - 1$, which gives error $(n - 1)/\varepsilon$; we call this the *standard Laplace mechanism*. (ii) *Groups*: if P and Q are joined by a path of h node neighbours in $\mathcal{F}$, then $\Pr[M(P) \in T] \leq e^{h\varepsilon} \Pr[M(Q) \in T]$. (iii) The following two-point bound, which we prove for completeness.

► **Lemma 7.** Let M be ε -node-private on $\mathcal{F}$, and let $P, Q \in \mathcal{F}$ be joined by a path of h node neighbours in $\mathcal{F}$ with $h\varepsilon \leq 1$. If $U(Q) - U(P) = \Delta > 0$, then $\mathbb{E}|M(P) - U(P)|$ or $\mathbb{E}|M(Q) - U(Q)|$ is at least $\Delta/(2(1 + e))$.

Proof. Let α be the larger of the two expected errors and $T = [U(P) + \Delta/2, \infty)$. By Markov’s inequality $\Pr[M(P) \in T] \leq 2\alpha/\Delta$ and $\Pr[M(Q) \notin T] \leq 2\alpha/\Delta$. By (ii), $1 - 2\alpha/\Delta \leq \Pr[M(Q) \in T] \leq e \cdot 2\alpha/\Delta$, so $\alpha \geq \Delta/(2(1 + e))$. ◀

4 The reduction

► **Lemma 8.** For every n , the sequence $k \mapsto \Lambda_k(n)/k$ is non-increasing. Hence $\Lambda_j(n) \leq (j/k)\Lambda_k(n)$ for $j \geq k$.

Proof. For a fixed P , $S_k(P)/k$ is the average of the k largest degrees (with zero padding), which does not increase when k grows. A pointwise maximum of non-increasing sequences is non-increasing. ◀

For a graph G and a real number $D \geq 0$, let

$$g_D(G) = \max \left\{ \sum_{e \in E(G)} x_e : 0 \leq x_e \leq 1, \sum_{e \ni v} x_e \leq D \text{ for every vertex } v \right\}.$$

This is the value of the flow statistic of [19], written as a fractional degree-constrained subgraph (Appendix A). We write $g_D(P)$ for $g_D(G(P))$.

► **Lemma 9.** For every real $D \geq 0$: (a) if P, Q are node neighbours then $|g_D(P) - g_D(Q)| \leq D$; (b) $U(P) - \text{tail}_D(P) \leq g_D(P) \leq U(P)$ for every P .

Proof. (a) Let $Q = P \cup \{q\}$. An optimal solution for Q , restricted to the edges not containing q , is feasible for P and loses at most $\sum_{e \ni q} x_e \leq D$. An optimal solution for P , extended by zeros, is feasible for Q . (b) The upper bound holds since $x_e \leq 1$. For the lower bound put $x_e = \min\{1, D/\deg u, D/\deg v\}$ for $e = uv$. At each vertex v the sum is at most $\deg v \cdot \min\{1, D/\deg v\} \leq D$. Each edge loses $1 - x_e \leq (1 - D/\deg u)_+ + (1 - D/\deg v)_+$, and summing over edges gives $\sum_v \deg v (1 - D/\deg v)_+ = \text{tail}_D(P)$. ◀

► **Lemma 10.** For every $P \in \mathcal{P}_n$ and real $D \geq 0$, $\text{tail}_D(P) \leq \max_{0 \leq j \leq n} (\Lambda_j(n) - jD)$, where $\Lambda_0 = 0$.

Proof. Let S be the set of points of degree larger than D and $j = |S|$. Then $\text{tail}_D(P) = \sum_{p \in S} \deg_P(p) - jD \leq S_j(P) - jD \leq \Lambda_j(n) - jD$. ◀

► **Theorem 1 (Reduction).** For every norm on $\mathbb{R}^2$, every $n \geq 1$ and every $\varepsilon \in (0, 1]$, with $k = \lceil 1/\varepsilon \rceil$,

$$\frac{1}{8(1+e)} \Lambda_k(n) \leq \text{err}^*(\varepsilon, n) \leq 2 \Lambda_k(n).$$

The upper bound is attained by an algorithm that is ε -node-private on all finite point sets. The same holds, with Λ_k defined accordingly, for every family of point sets closed under deleting points and every graph that is determined by pairs of points (for instance, any distance-bin graph).

Proof. *Upper bound.* Let $k = \lceil 1/\varepsilon \rceil$, $\lambda = \Lambda_k(n)$ and $D = \lambda/k$. The algorithm outputs $g_D(P) + Z$ with Z Laplace of scale D/ε ; it is ε -node-private on all finite sets by Lemma 9(a). For $j \geq k$, Lemma 8 gives $\Lambda_j(n) - jD \leq 0$, and for $j < k$ we have $\Lambda_j(n) - jD \leq \lambda$. By Lemmas 9 and 10(b), for every $P \in \mathcal{P}_n$,

$$\mathbb{E}|g_D(P) + Z - U(P)| \leq \text{tail}_D(P) + \mathbb{E}|Z| \leq \lambda + \frac{\lambda}{k\varepsilon} \leq 2\lambda.$$

Lower bound. Let $m = \min\{\lceil 1/\varepsilon \rceil, n\} \geq 1$ and choose $P^* \in \mathcal{P}_n$ with $S_m(P^*) = \Lambda_m(n)$. Let C consist of $\min\{m, |P^*|\}$ points of P^* of largest degree and $P_0 = P^* \setminus C$. Adding the points of C to P_0 one at a time gives a path of $|C| \leq m$ node neighbours inside $\mathcal{P}_n$. Every edge counted in $\sum_{c \in C} \deg(c) = \Lambda_m(n)$ is counted at most twice, so $U(P^*) - U(P_0) \geq \Lambda_m(n)/2$. Since $m\varepsilon \leq 1$, Lemma 7 shows that every ε -node-private algorithm has expected error at least $\Lambda_m(n)/(4(1+e))$ on P_0 or on P^* . Finally, if $m = n$ then $\Lambda_k(n) = \Lambda_m(n)$; otherwise $k \leq \lceil 1/\varepsilon \rceil + 1 \leq 2m$, and Lemma 8 gives $\Lambda_k(n) \leq 2\Lambda_m(n)$.

Other families. The upper bound used only Lemma 9, which holds for all graphs. The lower bound used only that P_0 and the intermediate sets belong to the family. Both hold for any family closed under deleting points and any graph in which adjacency of two points depends only on these two points. ◀

► **Remark 11.** The algorithm needs the value $\Lambda_k(n)$. If only an upper bound $\lambda \geq \Lambda_k(n)$ is known, the same proof with $D = \lambda/k$ gives error at most 2λ . For the Euclidean norm this yields error $O(n + n^{2/3}\varepsilon^{-2/3})$ from Theorem 3(2). Theorem 12 below removes the need for any bound at a logarithmic cost.

5 Top-degree mass and incidences

► **Theorem 2** (Top-degree mass and incidences). *For every norm on $\mathbb{R}^2$ and all integers $n \geq 2$ and $1 \leq k \leq n$,*

$$\max\left\{n-1, \frac{2ku(n)}{n}\right\} \leq \Lambda_k(n) \leq I(n, k),$$

where $u(n)$ is the maximum number of unit distances among n points in this norm. If $k \leq n/2$, then also $I(n, k) \leq 2\Lambda_k(n)$.

Proof. *Lower bounds.* Placing $n-1$ points on the unit circle around a further point gives $S_1 \geq n-1$. If $|P| = n$ and $U(P) = u(n)$, the average degree is $2u(n)/n$, and the k largest degrees have at least this average, so $S_k(P) \geq 2ku(n)/n$.

$\Lambda_k(n) \leq I(n, k)$. Let $P \in \mathcal{P}_n$ and let C be a set of $\min\{k, |P|\}$ points of P of largest degree. The unit circles $c + \mathbb{S}$, $c \in C$, are distinct, and $S_k(P) = \sum_{c \in C} |P \cap (c + \mathbb{S})| = I(P, \{c + \mathbb{S}\}_{c \in C}) \leq I(n, k)$.

$I(n, k) \leq 2\Lambda_k(n)$ for $k \leq n/2$. Let B be a set of at most n points and R a set of k centres. Split B into two parts of at most $\lceil n/2 \rceil$ points each, and let B' be a part that carries at least half of the incidences. The set $Q = B' \cup R$ has at most $\lceil n/2 \rceil + k \leq n$ points. Each $r \in R$ has $\deg_Q(r) \geq |B' \cap (r + \mathbb{S})|$, hence $\Lambda_k(n) \geq S_k(Q) \geq \sum_{r \in R} \deg_Q(r) \geq I(B, R)/2$. ◀

Lemma 10 also bounds the degree tail of every set by the top-degree mass. With Theorem 3(2) this gives, for every strictly convex norm,

$$\text{tail}_D(P) \leq \max_j (C(n^{2/3}j^{2/3} + n) - jD) = O(n^2/D^2 + n) \quad (P \in \mathcal{P}_n, D \geq 1). \quad (1)$$

Both terms are needed: a star with $2D$ leaves repeated $\lfloor n/(2D+1) \rfloor$ times has tail $\Omega(n)$, and by Theorem 3(3) the first term is attained for Valtr's norm. We note that (1) applies the incidence bound once, to the circles centred at the high-degree points; summing a bound on the number of j -rich points over all $j > D$ would lose a logarithmic factor.

6 Norms on the plane

► **Theorem 3** (Top-degree mass across norms). *There are absolute constants $C, c > 0$ such that for all $n \geq 8$ and $1 \leq k \leq n$:*

1. *If the unit circle contains a segment, then $\lfloor n/2 \rfloor \min\{k, n-1\} \leq \Lambda_k(n) \leq (n-1)k$.*
2. *If the norm is strictly convex, then $\Lambda_k(n) \leq C(n^{2/3}k^{2/3} + n)$.*
3. *For Valtr's norm $\|(x, y)\| = |y| + \sqrt{x^2 + y^2}$, we have $\Lambda_k(n) \geq c(n^{2/3}k^{2/3} + n)$.*
4. *For almost all norms, $n-1 \leq \Lambda_k(n) \leq 2n \log_2 n$.*

Proof of (1). Let the segment $\{v + tw : |t| \leq 1\}$ lie on $\mathbb{S}$. Put $m = \lfloor n/2 \rfloor$, $a_i = (i/m)w$ and $b_j = v + (j/m)w$ for $0 \leq i, j < m$. Then $b_j - a_i = v + \frac{j-i}{m}w$ lies on the segment, so all m^2 pairs $\{a_i, b_j\}$ are at unit distance. (The points are distinct, since a segment on $\mathbb{S}$ does not lie on a line through the origin.) Every point has degree at least m , and $2m \geq n-1$, so $S_k \geq m \min\{k, n-1\}$. The upper bound holds because every degree is at most $n-1$. ◀

Proof of (2). In a strictly convex norm, two distinct unit circles meet in at most two points, and two distinct points lie on at most two common unit circles (Lemma 17 in Appendix C). For such families, Székely's crossing-number argument [28], or the theorem of Pach and Sharir [21], gives $I(P, \Gamma) \leq C_0(|P|^{2/3}|\Gamma|^{2/3} + |P| + |\Gamma|)$; we include the argument in Appendix C. By Theorem 2, $\Lambda_k(n) \leq I(n, k) \leq C_0(n^{2/3}k^{2/3} + n + k)$, and $k \leq n$. ◀

Proof of (3). The norm $\|(x, y)\| = |y| + \sqrt{x^2 + y^2}$ is the sum of the Euclidean norm and a seminorm, so it is strictly convex [30, 27]. For a centre $c = (a, b)$ and a point $p = (x, y)$ with $y \geq b$, the equation $\|p - c\| = 1$ is equivalent to

$$y = b + \frac{1}{2}(1 - (x - a)^2), \quad |x - a| \leq 1.$$

The bijection $\Phi(x, y) = (x, y + x^2/2)$ maps this arc into the line $Y = aX + (b + \frac{1}{2} - \frac{a^2}{2})$. Conversely, every point (X, Y) with $|X - a| \leq 1$ on the line $Y = aX + \beta$ is the image of a point at unit distance from the centre $(a, \beta - \frac{1}{2} + \frac{a^2}{2})$.

Now use the grid construction for point–line incidences. For integers $A, B \geq 1$, the $2A^2B$ points (i, j) with $1 \leq i \leq A$, $1 \leq j \leq 2AB$ and the AB^2 lines $j = si + t$ with $1 \leq s \leq B$, $1 \leq t \leq AB$ have A^2B^2 incidences, since each line contains A of the points. Apply $(i, j) \mapsto (\frac{i}{2A}, \frac{j}{4AB})$: the lines get slopes $a = \frac{s}{2B} \in (0, \frac{1}{2}]$, and all points have $X \in (0, \frac{1}{2}]$, so $|X - a| \leq 1$. Applying Φ^{-1} to the points and taking the centres described above yields $2A^2B$ points and AB^2 distinct centres with at least A^2B^2 pairs at unit distance. Choosing A and B for given n and k (Appendix C) gives $I(\lfloor n/2 \rfloor, k) \geq n^{2/3}k^{2/3}/53$ for $\sqrt{n} \leq k \leq \lfloor n/2 \rfloor$. By Theorem 2 and monotonicity this gives (3), with $\Lambda_k(n) \geq n - 1$ for $k < \sqrt{n}$. ◀

Proof of (4). Alon, Bucić and Sauermaun [2] prove that for all norms on $\mathbb{R}^2$ outside a meagre set, every n points span at most $n \log_2 n$ unit distances. Then $\Lambda_k(n) \leq \Lambda_n(n) = 2u(n) \leq 2n \log_2 n$. The lower bound $n - 1$ holds in every norm. ◀

Proof of Corollary 4. Let $k = \lceil 1/\varepsilon \rceil \leq n - 1$ and apply Theorem 1. With a segment, $\Lambda_k(n) \geq \lfloor n/2 \rfloor / \varepsilon$, and the standard Laplace mechanism has error $(n - 1)/\varepsilon$. For strictly convex norms, (2) gives $2\Lambda_k(n) = O(n + n^{2/3}k^{2/3})$ and $k \leq 2/\varepsilon$; (3) gives the matching lower bound for Valtr’s norm. For almost all norms, (4) gives $\Omega(n)$ and $O(n \log n)$. ◀

► **Theorem 5 (Euclidean plane).** *There are absolute constants $C, c > 0$ such that for the Euclidean norm, all $n \geq 8$ and all $\varepsilon \in [1/n, 1]$,*

$$c \left(n + \frac{u(n)}{\varepsilon n} \right) \leq \text{err}^*(\varepsilon, n) \leq C \left(n + n^{2/3} \varepsilon^{-2/3} \right).$$

In particular $\text{err}^*(\varepsilon, n) = \Theta(n)$ for $\varepsilon \geq n^{-1/2}$. Moreover, the following are equivalent:

1. $u(n) \geq n^{1+\delta}$ for some $\delta > 0$ and infinitely many n ;
2. there are $\delta' > 0$ and privacy levels $\varepsilon_n \in (0, 1]$ such that $\text{err}^*(\varepsilon_n, n) \geq n^{1+\delta'}$ for infinitely many n .

By the 2026 results, both hold: for every $\eta \in [0, 0.014)$ there are infinitely many n with $\text{err}^*(n^{-1+\eta}, n) \geq cn^{1.014-\eta}$, while for almost all norms the error is $O(n \log n)$ at every ε .

Proof. Let $k = \lceil 1/\varepsilon \rceil$, so $1/\varepsilon \leq k \leq \min\{n, 2/\varepsilon\}$. By Theorems 1 and 2, $\text{err}^*(\varepsilon, n) \geq \frac{1}{8(1+\varepsilon)} \max\{n - 1, 2u(n)/(\varepsilon n)\}$. The upper bound follows from Theorems 1 and 3(2). For $\varepsilon \geq n^{-1/2}$ we have $n^{2/3} \varepsilon^{-2/3} \leq n$.

For the equivalence, if $u(n) \geq n^{1+\delta}$ then at $\varepsilon = 1/n$ the lower bound is at least $cu(n) \geq cn^{1+\delta}$, which exceeds $n^{1+\delta/2}$ for large n . Conversely, $\text{err}^*(\varepsilon, n) \leq 2\Lambda_k(n) \leq 2\Lambda_n(n) = 4u(n)$ for every ε , so a lower bound $n^{1+\delta'}$ on the error forces $u(n) \geq n^{1+\delta'}/4$. Finally, Sawin [24] gives $u(n) > n^{1.014}$ for infinitely many n ; for these n and $\varepsilon = n^{-1+\eta}$ the lower bound is at least $cn^{-\eta}u(n) \geq cn^{1.014-\eta}$. The comparison with almost all norms is Corollary 4. ◀

316 **What the separation does and does not say.** The separation in Theorem 5 appears only
 317 for $\varepsilon \leq n^{-1+0.014}$, that is, for very strong privacy. At such ε the optimal error is already
 318 comparable to the count itself, so we do not see it as a recommendation for practice. Its
 319 content is structural: the shape of the Euclidean rate curve near $\varepsilon = 1/n$ is decided exactly
 320 by whether $u(n)$ is polynomially superlinear, and the 2026 results settle this. For larger ε
 321 the curve is decided by $I(n, k)$ with $k < n$, where the gap in Theorem 5 is open.

322 7 Algorithms and other distance graphs

323 **An algorithm that does not need Λ_k .** The algorithm of Theorem 1 uses the unknown
 324 number $\Lambda_k(n)$. The following result, which combines the surrogate of [19] with the generalized
 325 exponential mechanism of Raskhodnikova and Smith [23], adapts to the input instead. It is
 326 not new as a method; we include it because its guarantee has a direct geometric reading.

327 **► Theorem 12.** *Let $K = 1 + \lceil \log_2 n \rceil$. There is an ε -node-private algorithm such that for*
 328 *every $P \in \mathcal{P}_n$ and $\beta \in (0, \frac{1}{2})$, with probability at least $1 - \beta$, its error is $O(S_t(P) + t)$ with*
 329 *$t = \lceil \log(K/\beta)/\varepsilon \rceil$. Consequently the error is $O(\log(K/\beta) (\Lambda_{\lceil 1/\varepsilon \rceil}(n) + 1/\varepsilon))$.*

330 The algorithm selects a threshold D from $\{1, 2, 4, \dots, 2^{K-1}\}$ privately and then releases
 331 $g_D(P)$ with Laplace noise; the proof is in Appendix B. The bound $S_t(P)$ is the number of
 332 incidences between P and the t richest unit circles centred in P . For instance, if every point
 333 of P has degree at most Δ , the error is $O(\Delta \log(K/\beta)/\varepsilon)$ regardless of n .

334 **Surrogates.** Theorem 1 needs real thresholds, because $\Lambda_k(n)/k$ need not be an integer; this
 335 is one reason to use the fractional statistic g_D . The integral statistic $f_D(G)$, the maximum
 336 number of edges of a subgraph of maximum degree at most D , satisfies the same two
 337 properties for integer D and can be computed by b -matching [25, 15]. With D rounded up
 338 it gives Theorem 1 up to an additive $1/\varepsilon$ (Appendix A). Both statistics are computable in
 339 polynomial time from $G(P)$.

340 **Other distance graphs.** Theorem 1 applies to every family of point sets that is closed
 341 under deleting points. Two examples are worked out in Appendix E. If the points lie on
 342 a grid $\frac{1}{q}\mathbb{Z}^2$ and the target distance r satisfies $qr \in \mathbb{Z}$, every degree is at most the number
 343 Δ of representations of $(qr)^2$ as a sum of two squares, and the error is $\Theta(\Delta/\varepsilon)$ whenever
 344 $n \geq \lceil 1/\varepsilon \rceil (\Delta + 1)$. If the points are δ -separated and adjacency means a distance in $[r - t, r + t]$,
 345 every degree is $O((t + \delta)(r + t + \delta)/\delta^2)$; in this case releasing g_D for D equal to this bound
 346 is private on all inputs and exact on separated inputs. Finally, Appendix D treats a class of
 347 point sets defined by an upper bound on the number of j -rich points for every j ; this class
 348 is not closed under deletion, so Theorem 1 does not apply, and we give separate matching
 349 bounds.

350 8 Open problems

351 **The Euclidean window.** For $\alpha \in [\frac{1}{2}, 1]$ let $\gamma(\alpha) = \limsup_{n \rightarrow \infty} \log \Lambda_{\lceil n^\alpha \rceil}(n) / \log n$ for the
 352 Euclidean norm, and let $1 + \delta^* = \gamma(1)$ be the unit-distance exponent. Theorems 2 and 3 give

$$353 \quad \max\{1, \alpha + \delta^*\} \leq \gamma(\alpha) \leq \frac{2+2\alpha}{3},$$

354 with $\delta^* \geq 0.014$, and for Valtr's norm the upper bound is attained. Is the lower bound the
 355 truth? In words: can a few unit circles be much richer than average while the unit-distance

graph as a whole stays sparse? By Theorem 2 this is a question about incidences between n points and k unit circles. We are not aware of results on this unbalanced version beyond the bounds above.

Instance-wise bounds. Theorem 1 concerns the worst case over $\mathcal{P}_n$. A local version would compare an algorithm, at each input P , with the best algorithm on a small neighbourhood of P . Deleting points is controlled by $S_t(P)$, as in Theorem 12; adding points brings in the external quantity $\max_{c \notin P} |P \cap (c + \mathbb{S})|$ and its analogues for several inserted points.

Higher dimensions. In $\mathbb{R}^3$, $u(n)$ lies between $n^{4/3} \log \log n$ [5] and $n^{295/197+o(1)}$ [31], so the window is wider and both ends are open. In $\mathbb{R}^d$ with $d \geq 4$, points on two orthogonal circles of radius $1/\sqrt{2}$ form a complete bipartite unit-distance graph, as in Theorem 3(1), and the standard Laplace mechanism is optimal.

References

- 1 Noga Alon, Thomas F. Bloom, W. T. Gowers, Daniel Litt, Will Sawin, Arul Shankar, Jacob Tsimmerman, Victor Wang, and Melanie Matchett Wood. Remarks on the disproof of the unit distance conjecture, 2026. URL: <https://doi.org/10.48550/arXiv.2605.20695>, arXiv:2605.20695.
- 2 Noga Alon, Matija Bucić, and Lisa Sauermann. Unit and distinct distances in typical norms, 2023. URL: <https://arxiv.org/abs/2302.09058>, arXiv:2302.09058.
- 3 Hilal Asi and John C. Duchi. Instance-optimality in differential privacy via approximate inverse sensitivity mechanisms. In *Advances in Neural Information Processing Systems 33 (NeurIPS)*, pages 14106–14117, 2020. URL: <https://proceedings.neurips.cc/paper/2020/hash/a267f936e54d7c10a2bb70dbe6ad7a89-Abstract.html>.
- 4 Jeremiah Blocki, Avrim Blum, Anupam Datta, and Or Sheffet. Differentially private data analysis of social networks via restricted sensitivity. In *Proceedings of the 4th Conference on Innovations in Theoretical Computer Science (ITCS)*, pages 87–96. ACM, 2013. doi:10.1145/2422436.2422449.
- 5 Peter Brass, William O. J. Moser, and János Pach. *Research Problems in Discrete Geometry*. Springer, 2005. doi:10.1007/0-387-29929-7.
- 6 Hongjie Chen, Jingqiu Ding, Tommaso d’Orsi, Yiding Hua, Chih-Hung Liu, and David Steurer. Private graphon estimation via sum-of-squares. In *Proceedings of the 56th Annual ACM Symposium on Theory of Computing (STOC)*, pages 172–182. ACM, 2024. doi:10.1145/3618260.3649643.
- 7 Kenneth L. Clarkson, Herbert Edelsbrunner, Leonidas J. Guibas, Micha Sharir, and Emo Welzl. Combinatorial complexity bounds for arrangements of curves and spheres. *Discrete & Computational Geometry*, 5:99–160, 1990. doi:10.1007/BF02187783.
- 8 Rachel Cummings and David Durfee. Individual sensitivity preprocessing for data privacy. In *Proceedings of the 2020 ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 528–547. SIAM, 2020. doi:10.1137/1.9781611975994.32.
- 9 Wei Dong, Juanru Fang, Ke Yi, Yuchao Tao, and Ashwin Machanavajjhala. R2T: Instance-optimal truncation for differentially private query evaluation with foreign keys. In *Proceedings of the 2022 International Conference on Management of Data (SIGMOD)*. ACM, 2022.
- 10 Cynthia Dwork, Frank McSherry, Kobbi Nissim, and Adam D. Smith. Calibrating noise to sensitivity in private data analysis. In *Theory of Cryptography*, volume 3876 of *Lecture Notes in Computer Science*, pages 265–284. Springer, 2006. doi:10.1007/11681878_14.
- 11 Cynthia Dwork and Aaron Roth. The algorithmic foundations of differential privacy. *Foundations and Trends in Theoretical Computer Science*, 9(3–4):211–407, 2014. doi:10.1561/04000000042.

- 403 12 Michael T. M. Emmerich. Optimizing explicit unit-distance lower-bound certificates, 2026.
404 URL: <https://doi.org/10.48550/arXiv.2606.03419>, arXiv:2606.03419.
- 405 13 Paul Erdős. On sets of distances of n points. *American Mathematical Monthly*, 53(5):248–250,
406 1946. URL: <https://www.jstor.org/stable/2305092>.
- 407 14 Juanru Fang, Wei Dong, and Ke Yi. Shifted inverse: A general mechanism for monotonic
408 functions under user differential privacy. In *Proceedings of the 2022 ACM SIGSAC Conference*
409 *on Computer and Communications Security (CCS)*, pages 1009–1022. ACM, 2022. doi:
410 10.1145/3548606.3560567.
- 411 15 Harold N. Gabow. Data structures for weighted matching and extensions to b -matching and
412 f -factors. *ACM Transactions on Algorithms*, 14(3):39:1–39:80, 2018. doi:10.1145/3183369.
- 413 16 G. H. Hardy and E. M. Wright. *An Introduction to the Theory of Numbers*. Oxford University
414 Press, 6 edition, 2008. Revised by D. R. Heath-Brown and J. H. Silverman.
- 415 17 Palak Jain, Adam D. Smith, and Connor Wagaman. Time-aware projections: Truly node-
416 private graph statistics under continual observation. In *2024 IEEE Symposium on Security and*
417 *Privacy (SP)*, pages 127–145. IEEE, 2024. URL: <https://doi.org/10.48550/arXiv.2403.04630>,
418 arXiv:2403.04630.
- 419 18 Iden Kalemaj, Sofya Raskhodnikova, Adam D. Smith, and Charalampos E. Tsourakakis.
420 Node-differentially private estimation of the number of connected components. In *Proceedings*
421 *of the 42nd ACM SIGMOD-SIGACT-SIGAI Symposium on Principles of Database Systems*
422 *(PODS)*, pages 183–194. ACM, 2023. doi:10.1145/3584372.3588671.
- 423 19 Shiva Prasad Kasiviswanathan, Kobbi Nissim, Sofya Raskhodnikova, and Adam D. Smith. Ana-
424 lyzing graphs with node differential privacy. In *Theory of Cryptography*, volume 7785 of *Lecture*
425 *Notes in Computer Science*, pages 457–476. Springer, 2013. doi:10.1007/978-3-642-36594-2_
426 26.
- 427 20 OpenAI. An OpenAI model has disproved a central conjecture in discrete geometry, 2026.
428 URL: <https://openai.com/index/model-disproves-discrete-geometry-conjecture/>.
- 429 21 János Pach and Micha Sharir. On the number of incidences between points and curves. *Com-*
430 *binatorics, Probability and Computing*, 7(1):121–127, 1998. doi:10.1017/S0963548397003192.
- 431 22 Sofya Raskhodnikova, Adam Smith, Connor Wagaman, and Anatoly Zavyalov. Local node
432 differential privacy, 2026. URL: <https://arxiv.org/abs/2602.15802>, arXiv:2602.15802.
- 433 23 Sofya Raskhodnikova and Adam D. Smith. Lipschitz extensions for node-private graph
434 statistics and the generalized exponential mechanism. In *57th Annual IEEE Symposium on*
435 *Foundations of Computer Science (FOCS)*, pages 495–504. IEEE Computer Society, 2016.
436 doi:10.1109/FOCS.2016.60.
- 437 24 Will Sawin. An explicit lower bound for the unit distance problem, 2026. URL: <https://doi.org/10.48550/arXiv.2605.20579>, arXiv:2605.20579.
- 439 25 Alexander Schrijver. *Combinatorial Optimization: Polyhedra and Efficiency*, volume 24 of
440 *Algorithms and Combinatorics*. Springer, 2003.
- 441 26 Joel Spencer, Endre Szemerédi, and William T. Trotter, Jr. Unit distances in the Euclidean
442 plane. In Béla Bollobás, editor, *Graph Theory and Combinatorics*, pages 293–303. Academic
443 Press, 1984.
- 444 27 Konrad J. Swanepoel. Combinatorial distance geometry in normed spaces. In *New Trends in*
445 *Intuitive Geometry*, volume 27 of *Bolyai Society Mathematical Studies*, pages 407–458. Springer,
446 2018. URL: <https://arxiv.org/abs/1702.00066>, arXiv:1702.00066.
- 447 28 László A. Székely. Crossing numbers and hard Erdős problems in discrete geometry. *Combina-*
448 *torics, Probability and Computing*, 6(3):353–358, 1997.
- 449 29 Endre Szemerédi and William T. Trotter, Jr. Extremal problems in discrete geometry.
450 *Combinatorica*, 3:381–392, 1983. doi:10.1007/BF02579194.
- 451 30 Pavel Valtr. Strictly convex norms allowing many unit distances and related touching ques-
452 tions, 2005. Manuscript. The construction is reproduced in the survey by K. J. Swanepoel,
453 *Combinatorial distance geometry in normed spaces*, 2018.

- 454 31 Joshua Zahl. Breaking the $3/2$ barrier for unit distances in three dimensions. *International*
455 *Mathematics Research Notices*, 2019. [arXiv:1706.05118](#).

Appendix

A Surrogate statistics

The flow statistic. Kasiviswanathan, Nissim, Raskhodnikova and Smith [19] define the following network for a graph $G = (V, E)$ and a capacity $D \geq 0$. Take a source, a sink, and two copies v_ℓ, v_r of each vertex. Add arcs of capacity D from the source to each v_ℓ and from each v_r to the sink, and for each edge uv add arcs (u_ℓ, v_r) and (v_ℓ, u_r) of capacity 1. Let $\phi_D(G)$ be the maximum flow value. The definition makes sense for real D .

► **Proposition 13.** *For every graph G and real $D \geq 0$, $\phi_D(G)/2 = g_D(G)$.*

Proof. Given a feasible x for g_D , send x_{uv} units along both arcs of each edge. This is a feasible flow of value $2 \sum_e x_e$. Conversely, from a feasible flow y set $x_{uv} = (y_{u_\ell v_r} + y_{v_\ell u_r})/2$. Then $0 \leq x_e \leq 1$, and $\sum_{e \ni v} x_e$ is half the flow through v_ℓ plus half the flow through v_r , which is at most D . The objective is half the flow value. ◀

Hence Lemma 9 is a property of the statistic of [19]; our only additions are the explicit fractional solution in part (b), which gives the bias bound for every real D , and the remark that real thresholds are needed in Theorem 1.

The integral statistic. For an integer $D \geq 0$ let $f_D(G)$ be the maximum number of edges in a subgraph of G with maximum degree at most D .

► **Proposition 14.** *For integer $D \geq 0$: (a) $|f_D(G) - f_D(G - v)| \leq D$; (b) $|E| - \text{tail}_D(G) \leq f_D(G) \leq g_D(G) \leq |E|$, and both inner inequalities can be strict; (c) f_D and g_D can be computed in polynomial time.*

Proof. (a) An optimal subgraph for G has at most D edges at v ; removing them gives a feasible subgraph of $G - v$. Every feasible subgraph of $G - v$ is feasible for G . (b) Repeatedly delete an edge at a vertex of current degree above D ; each deletion lowers $\sum_v (\deg v - D)_+$ by at least one, so at most $\text{tail}_D(G)$ edges are deleted. Integral solutions are feasible for g_D . For a triangle and $D = 1$, $f_1 = 1 < 3/2 = g_1$; for a star with more than D leaves, $g_D = D < |E|$. (c) g_D is a maximum flow; f_D is a maximum-cardinality simple b -matching with $b \equiv D$ [25, 15]. ◀

With $D' = \lceil \Lambda_k(n)/k \rceil$ in place of D in the proof of Theorem 1, the bias is at most $\text{tail}_{D'} \leq \text{tail}_D \leq \Lambda_k(n)$ and the noise is at most $(D + 1)/\varepsilon \leq \Lambda_k(n) + 1/\varepsilon$, so $f_{D'}$ gives error at most $2\Lambda_k(n) + 1/\varepsilon$.

B The adaptive algorithm

Let $\mathcal{D} = \{1, 2, 4, \dots, 2^{K-1}\}$ with $K = 1 + \lceil \log_2 n \rceil$, so $\max \mathcal{D} \geq n - 1$. Fix $\beta \in (0, \frac{1}{2})$, split $\varepsilon = \varepsilon_1 + \varepsilon_2$ with $\varepsilon_1 = \varepsilon_2 = \varepsilon/2$, and let $a = \log(2/\beta)$.

Algorithm. Run the generalized exponential mechanism of Raskhodnikova and Smith [23, Theorem 1.4] with privacy parameter ε_1 on the scores $q_D(P) = -g_D(P) + aD/\varepsilon_2$, $D \in \mathcal{D}$, where score q_D is declared to have sensitivity D ; let $\hat{D}$ be the output. Release $g_{\hat{D}}(P) + Z$ with Z Laplace of scale $\hat{D}/\varepsilon_2$.

► **Lemma 15.** *The algorithm is ε -node-private. With probability at least $1 - \beta$ its error is at most $\min_{D \in \mathcal{D}} \{\text{tail}_D(P) + aD/\varepsilon_2 + 4D \log(2K/\beta)/\varepsilon_1\}$.*

Proof. Each score q_D has sensitivity at most D by Lemma 9(a), since aD/ε_2 does not depend on P . The selection step is ε_1 -node-private by [23, Theorem 1.4], the release step is ε_2 -node-private given $\widehat{D}$, and the composition is ε -node-private. By [23, Theorem 1.4], with probability at least $1 - \beta/2$, $q_{\widehat{D}}(P) \leq \min_D \{q_D(P) + 4D \log(2K/\beta)/\varepsilon_1\}$. With probability at least $1 - \beta/2$, $|Z| \leq a\widehat{D}/\varepsilon_2$. On both events,

$$|g_{\widehat{D}}(P) + Z - U(P)| \leq U(P) - g_{\widehat{D}}(P) + a\widehat{D}/\varepsilon_2 = U(P) + q_{\widehat{D}}(P),$$

and $U(P) + q_D(P) \leq \text{tail}_D(P) + aD/\varepsilon_2$ by Lemma 9(b). $\blacktriangleleft$

► **Lemma 16.** *Let $d_{(1)} \geq d_{(2)} \geq \dots$ be the degrees of $G(P)$, padded with zeros. For every integer $t \geq 1$, $\min_{D \geq 0} \{\text{tail}_D(P) + tD\} = S_t(P)$, and $\min_{D \in \mathcal{D}} \{\text{tail}_D(P) + tD\} \leq 2S_t(P) + t$.*

Proof. For every $D \geq 0$, $\text{tail}_D(P) + tD \geq \sum_{i \leq t} (d_{(i)} - D) + tD = S_t(P)$, with equality at $D = d_{(t+1)}$. For the grid, if $d_{(t+1)} = 0$ take $D = 1$: only the top t points have positive degree, so $\text{tail}_1(P) \leq S_t(P)$. Otherwise take $D \in \mathcal{D}$ with $d_{(t+1)} \leq D \leq 2d_{(t+1)}$; then $\text{tail}_D(P) + tD \leq S_t(P) - t d_{(t+1)} + 2t d_{(t+1)} \leq 2S_t(P)$. $\blacktriangleleft$

Proof of Theorem 12. By Lemma 15, the error is at most $\min_{D \in \mathcal{D}} \{\text{tail}_D(P) + t'D\}$ with $t' = 2a/\varepsilon + 8 \log(2K/\beta)/\varepsilon = O(t)$. By Lemma 16 and since $S_{t'} \leq \lceil t'/t \rceil S_t$, this is $O(S_t(P) + t)$. For the second statement let $k = \lceil 1/\varepsilon \rceil$. Then $S_t(P) \leq \Lambda_{\max\{t,k\}}(n) \leq (t/k + 1)\Lambda_k(n)$ by Lemma 8, and $t/k \leq \log(K/\beta) + 1$. $\blacktriangleleft$

C Deferred geometric proofs

► **Lemma 17.** *Let $\mathbb{S}$ be the unit circle of a strictly convex norm. Two distinct translates of $\mathbb{S}$ meet in at most two points, and two distinct points lie on at most two common translates of $\mathbb{S}$.*

Proof. Let K be the unit ball and $v \neq 0$. Choose coordinates (s, τ) with τ measured along v . Each line $\{s = s_0\}$ that meets the interior of K meets K in a segment $[\tau_-(s_0), \tau_+(s_0)]$, where τ_- is convex and τ_+ is concave; at the two extreme values of s the intersection is a single point, by strict convexity. The boundary of $K + v$ on the same line is $\{\tau_-(s_0) + |v|, \tau_+(s_0) + |v|\}$. Since $\tau_- \leq \tau_+$ and $|v| > 0$, a common boundary point must satisfy $\tau_+(s_0) = \tau_-(s_0) + |v|$, that is, $h(s_0) = |v|$ for $h = \tau_+ - \tau_-$. Strict convexity of K means that the graphs of $\tau_{\pm}$ contain no segment, so h is strictly concave, and it vanishes at the extreme values of s . A strictly concave function takes a positive value at most twice, which proves the first claim. For the second, $p, q \in c + \mathbb{S}$ means $c \in (p + \mathbb{S}) \cap (q + \mathbb{S})$ because $\mathbb{S} = -\mathbb{S}$, and these two translates are distinct. $\blacktriangleleft$

► **Lemma 18.** *There is an absolute constant C_0 such that for every strictly convex norm, every finite set P and every finite family Γ of distinct unit circles, $I(P, \Gamma) \leq C_0(|P|^{2/3}|\Gamma|^{2/3} + |P| + |\Gamma|)$.*

Proof. This is the argument of Székely [28]. Let $N = |P|$ and $m = |\Gamma|$. Draw a multigraph H on P : for every $\gamma \in \Gamma$ containing $j \geq 2$ points of P , join consecutive points along γ by the j arcs between them. Then H has at least $I(P, \Gamma) - m$ edges. By Lemma 17, two points are joined by arcs of at most two circles, and each circle contributes at most two such arcs, so every edge has multiplicity at most 4. Two arcs of the same circle do not cross, and two circles meet in at most two points, so the drawing has at most m^2 crossings. The crossing lemma for multigraphs with bounded multiplicity [28] states that a multigraph

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with N vertices, e edges and multiplicity at most μ has crossing number at least $c_1 e^3 / (\mu N^2)$ whenever $e \geq c_2 \mu N$, for absolute constants $c_1, c_2 > 0$. Hence either $I(P, \Gamma) - m < 4c_2 N$ or $m^2 \geq c_1 (I(P, \Gamma) - m)^3 / (4N^2)$, and in both cases the claim follows. $\blacktriangleleft$

Choice of parameters in Theorem 3(3). Let $n \geq 8$, $m = \lfloor n/2 \rfloor \geq 4$ and $\sqrt{n} \leq k \leq m$. Put $A_0 = (m/2)^{2/3} k^{-1/3}$ and $B_0 = k^{2/3} (m/2)^{-1/3}$, so that $2A_0^2 B_0 = m$ and $A_0 B_0^2 = k$. Since $k \leq m$ and $m \geq 4$ we have $(m/2)^2 \geq k$, so $A_0 \geq 1$; since $k^2 \geq n \geq m/2$ we have $B_0 \geq 1$. Let $A = \lfloor A_0 \rfloor \geq A_0/2$ and $B = \lfloor B_0 \rfloor \geq B_0/2$. The construction in the proof of Theorem 3(3) uses at most m points and at most k centres and has at least $A^2 B^2 \geq A_0^2 B_0^2 / 16 = (mk/2)^{2/3} / 16$ incidences. Since $m \geq n/3$, this is at least $(nk/6)^{2/3} / 16 \geq n^{2/3} k^{2/3} / 53$. By Theorem 2 (the set Q in its proof has at most $m + k \leq n$ points), $\Lambda_k(n) \geq n^{2/3} k^{2/3} / 53$. For $m < k \leq n$, monotonicity gives $\Lambda_k(n) \geq \Lambda_m(n) \geq n^{2/3} m^{2/3} / 53 \geq n^{2/3} k^{2/3} / 111$, using $m \geq n/3 \geq k/3$ and $m \geq \sqrt{n}$. For $k < \sqrt{n}$, $\Lambda_k(n) \geq n - 1 \geq n/2 \geq n^{2/3} k^{2/3} / 2$. Hence (3) holds with $c = 1/222$. $\blacktriangleleft$

D A class defined by rich points

In this appendix the norm is Euclidean. For $j \geq 1$ let $R_j(P)$ be the number of points of P of degree at least j . For $C \geq 1$ and $D_0 \geq 1$, call P (C, D_0) -regular if $R_j(P) \leq C|P|^2/j^3$ for every integer $j > D_0$. Deleting a point can violate this condition, so the class is not closed under deletion, and Theorem 1 does not apply. The bound $R_j(P) = O(n^2/j^3 + n/j)$ holds for every set by the incidence bound, so the condition removes exactly the term n/j coming from many stars of degree about j .

► **Proposition 19.** *If P is (C, D_0) -regular, $|P| = n$, and $D \geq D_0$, then $\text{tail}_D(P) = \sum_{j>D} R_j(P) = O(Cn^2/D^2)$. Hence the algorithm $g_D(P) + \text{Lap}(D/\varepsilon)$ with $D \asymp (C\varepsilon n^2)^{1/3} \geq D_0$ has expected error $O(C^{1/3} n^{2/3} \varepsilon^{-2/3})$, and the algorithm of Theorem 12 has error $O(C^{1/3} n^{2/3} (\log(K/\beta)/\varepsilon)^{2/3})$ with probability $1 - \beta$ whenever $D_0 \leq (Cn^2 \varepsilon / \log(K/\beta))^{1/3}$.*

Proof. $\text{tail}_D(P) = \sum_{j>D} R_j(P) \leq Cn^2 \sum_{j>D} j^{-3} = O(Cn^2/D^2)$. The first algorithm is private on all inputs by Lemma 9(a), and its error is $O(D/\varepsilon + Cn^2/D^2)$. For the second, Lemma 15 bounds the error by $O(\min_{D \in \mathcal{D}, D \geq D_0} \{Cn^2/D^2 + D \log(K/\beta)/\varepsilon\})$; optimise over D . $\blacktriangleleft$

► **Proposition 20.** *There are absolute constants $a, b > 0$ and n_0 such that for $n \geq n_0$, $1 \leq C \leq n$, $D_0 \geq 1$ and $aC^{1/2} n^{-1/2} \leq \varepsilon \leq 1$, every ε -node-private algorithm has expected error at least $bC^{1/3} n^{2/3} \varepsilon^{-2/3}$ on some (C, D_0) -regular set of at most n points.*

Proof. Let $h = \max\{1, \lfloor 1/(4\varepsilon) \rfloor\}$, so $1/(8\varepsilon) \leq h \leq 1/\varepsilon$. Choose a large and $\alpha \leq \frac{1}{2}$ small with $2\alpha a^{-2/3} \leq \frac{1}{2}$, and set $L = \lfloor \alpha(Cn^2/h)^{1/3} \rfloor$. Since $(Cn^2/h)^{1/3} \geq (Cn^2 \varepsilon)^{1/3} \geq a^{1/3} C^{1/2} n^{1/2}$, for $n \geq n_0$ we get $L \geq \frac{\alpha}{2} (Cn^2/h)^{1/3} \geq 2$ and $h(L+1) \leq 2\alpha C^{1/3} n^{2/3} \varepsilon^{-2/3} \leq n/2$. Take h far-apart gadgets, each consisting of a potential centre c_i and L points on its unit circle, with angles chosen so that no two of these L points are at unit distance, together with $n - h(L+1)$ isolated padding points. Let P_i contain all non-centre points and $c_1, \dots, c_i$. Then $|P_h| = n$, $|P_i| \geq n/2$, consecutive sets are node neighbours, and $U(P_h) - U(P_0) = hL$. For $2 \leq j \leq L$ the points of degree at least j are the inserted centres, so $R_j(P_i) \leq h \leq C|P_i|^2/L^3 \leq C|P_i|^2/j^3$ by the choice of α , and $R_j(P_i) = 0$ for $j > L$. Thus every P_i is (C, D_0) -regular. Since $h\varepsilon \leq 1$, Lemma 7 gives error at least $hL/(2(1+e)) = \Omega(C^{1/3} n^{2/3} h^{2/3}) = \Omega(C^{1/3} n^{2/3} \varepsilon^{-2/3})$. $\blacktriangleleft$

The class is useful as a conditional statement: it describes inputs on which the rich points are spread over many degree levels. We do not know a natural family of point sets with

many unit distances that satisfies the condition and is not built from stars, and we leave this as a question.

E Grid points and separated sets

Grid points. Let $P \subset \frac{1}{q}\mathbb{Z}^2$ and let the target distance r satisfy $qr \in \mathbb{Z}$. Points of $\frac{1}{q}\mathbb{Z}^2$ at distance r from a grid point correspond to integer solutions of $x^2 + y^2 = (qr)^2$, so every degree is at most $\Delta = r_2((qr)^2)$, the number of such representations, and $\Delta \leq (qr)^{O(1/\log \log(qr))}$ [16]. Grid point sets form a family closed under deletion, with $\Lambda_k \leq k\Delta$. A lattice point together with the Δ grid points on its circle shows $\Lambda_k \geq k\Delta$ when $n \geq k(\Delta + 1)$ (use k far-apart copies). By Theorem 1 the error is $\Theta(\Delta/\varepsilon)$ when $n \geq \lceil 1/\varepsilon \rceil(\Delta + 1)$; the Laplace mechanism applied to U itself achieves the upper bound. If arbitrary inputs are first rounded to a fixed public grid, neighbours are mapped to equal or neighbouring grid sets, so the rounded statistic inherits privacy.

Separated sets and annuli. Fix $\delta > 0$, $r > 0$, $t \geq 0$, and let two points be adjacent when their distance lies in $[r - t, r + t]$. Consider the family of δ -separated sets, in which every two points have distance at least δ ; it is closed under deletion. The disks of radius $\delta/2$ around the neighbours of a point p are disjoint and lie in the annulus around p with radii $\max\{0, r - t - \delta/2\}$ and $r + t + \delta/2$. If the inner radius is positive, this annulus has area $2\pi r(2t + \delta)$; otherwise $r \leq t + \delta/2$ and the area is at most $\pi(2t + \delta)^2$. In both cases the area is at most $\pi(2t + \delta)(2r + 2t + \delta)$, so every degree is at most

$$\Delta_{\text{sep}} = \frac{4(2t + \delta)(2r + 2t + \delta)}{\delta^2} = O\left(\frac{rt + t^2}{\delta^2} + \frac{r + t}{\delta} + 1\right).$$

For $t \leq r$ this is $O(rt/\delta^2 + r/\delta + 1)$. For $t > r$ the term t^2/δ^2 cannot be dropped: then every point within distance $r + t$ of p is a neighbour of p , and a disk of radius $t \geq \delta$ contains $\Omega(t^2/\delta^2)$ points of a δ -separated grid. The algorithm $g_D(P) + \text{Lap}(D/\varepsilon)$ with $D = \Delta_{\text{sep}}$ is ε -node-private on *all* finite sets by Lemma 9(a); on δ -separated sets every degree is at most D , so $g_D(P) = U(P)$ by Lemma 9(b) and the only error is the noise, $\Delta_{\text{sep}}/\varepsilon$ in expectation. The same argument applies to threshold graphs (distance at most r), with degree $O(r^2/\delta^2 + 1)$.

Several distance bins. For a histogram of distances with bins $b_1, \dots, b_L$, adding a point changes all bins at once. A simple private release splits $\varepsilon = \varepsilon_1 + \dots + \varepsilon_L$ and runs the one-bin algorithm with parameter ε_i in bin i .